



Flow simulation around bluff body: $k-\omega$ turbulence model

Computational Fluid Dynamics and Wind Engineering

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July 15, 2026



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Project Overview & Objectives

1 Initial Problem

Flow Configuration:

- Two-dimensional, incompressible flow past a sharp-edged square cylinder in a subcritical regime.

Physical Challenge:

- Accurately predicting:
 - Shear layer separation
 - Flow reattachment
 - Unsteady wake dynamics (vortex shedding)

Project Objective:

- Compare the performance of two RANS turbulence models:
 - $k - \varepsilon$ **RNG**
 - $k - \omega$ **Standard**



Flow Configuration & Boundary Setup

1 Initial Problem

Flow Properties:

- Unsteady, 2D flow past a square cylinder (reference dimension B).
- Subcritical regime: $Re \approx 2.2 \times 10^4$.
- Made dimensionless using $\rho = 1 \text{ kg/m}^3$, $\nu = 4.545 \times 10^{-5} \text{ m}^2/\text{s}$, and U_∞ .

Boundary Conditions:

- **Inlet:** Velocity-inlet, Turbulence Intensity = 2%, Length Scale = 0.5.
- **Outlet:** Dirichlet condition on pressure ($p_{rel} = 0$).
- **Upper/Lower Limits:** Periodic conditions.
- **Cylinder Surface:** Smooth edge (roughness = 0), no-slip ($u = v = 0$).

Initialization:

- Uniform initial condition matching inlet values across the entire domain.



Spatial Discretization

1 Initial Problem

Domain Partitioning:

- The computational domain is divided into distinct macro-zones.

Mesh Strategy:

- **Outer Area (EUL Region):** Mapped meshing approach;
- **Inner Area (ALE Region):** Unstructured mesh of triangular cells;
- **Grid Boundary Layer:** Finer layer of quadrilateral cells around the obstacle walls.



Turbulence Modeling: RANS approach

1 Initial Problem

RANS Approach:

- Non-scale resolving models;
- The equation contains a new term: Reynolds tensor $R_{ij} = \overline{u'_i u'_j}$.

The Closure Problem:

- Boussinesq hypothesis: it introduces the turbulent viscosity
$$\nu_t = C_\mu \frac{k^2}{\varepsilon} \implies \overline{u'_i u'_j} = 2\nu_t S_{ij};$$
- Requires transport equations for k and ε .



Turbulence Modeling: The $k - \varepsilon$ RNG Model

1 Initial Problem

- **Transport Equations for k and ε**

$$\frac{\partial k}{\partial t} + \frac{\partial \bar{u}_j k}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\left(\frac{\nu_t}{\sigma_k} + \nu \right) \frac{\partial k}{\partial x_j} \right] + P_k - \varepsilon$$

$$\frac{\partial \varepsilon}{\partial t} + \frac{\partial \bar{u}_j \varepsilon}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\left(\frac{\nu_t}{\sigma_\varepsilon} + \nu \right) \frac{\partial \varepsilon}{\partial x_j} \right] + \frac{\varepsilon}{k} C_{\varepsilon 1} P_k - \frac{\varepsilon}{k} C_{\varepsilon 2} \varepsilon$$

- **Standard Coefficients**

— 5 empirical parameters for dimensional consistency: C_μ , $C_{\varepsilon 1}$, $C_{\varepsilon 2}$, σ_k , σ_ε .

- **RNG Variant**

— Renormalization of the coefficients;
— Dynamic coefficient $C_{\varepsilon 2}$ instead of a constant value.

- **Result:**

— Correction for k and ν_t overestimation in stagnation regions.



Turbulence Modeling: Standard $k - \omega$ Model

1 Initial Problem

- **Specific Dissipation Rate ω**

$$\omega \propto \frac{\varepsilon}{k} \quad \text{dimensions: } [s^{-1}]$$

- **Transport Equations for ω**

$$\frac{\partial \omega}{\partial t} + \frac{\partial \bar{u}_j \omega}{\partial x_j} = \frac{\partial}{\partial x_j} \left[\left(\frac{\nu_t}{\sigma_\omega} + \nu \right) \frac{\partial \omega}{\partial x_j} \right] + \underbrace{\frac{\omega}{k} C_{\omega 1} P_k}_{P_\omega} - \underbrace{\frac{\omega}{k} C_{\omega 2} \varepsilon}_{D_\omega}$$

- **Empirical Coefficients**

— 5 empirical parameters for dimensional consistency: σ_k , σ_ω , $C_{\omega 1}$, $C_{\omega 2}$, β^*

- **Main Benefits (vs. $k - \varepsilon$)**

— P_ω and D_ω do not depend on k ;
— In the stagnation region there is no overestimation of k .



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► Initializing the Simulation

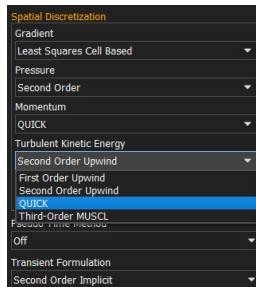
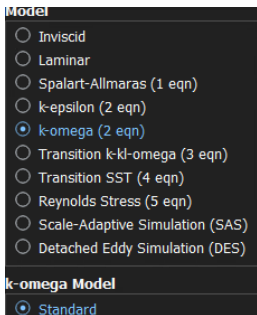
► Results



Changing Turbulence Model

2 Initializing the Simulation

We need to change the turbulence model from $k - \varepsilon$ RNG to $k - \omega$ standard.
Then we re-set the correct discretization scheme for k .

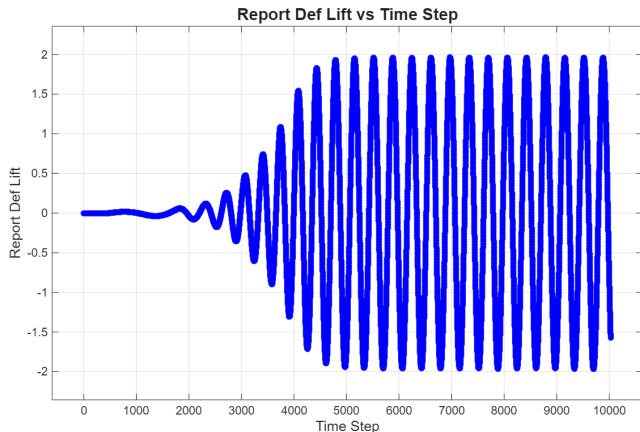




First Simulation

2 Initializing the Simulation

The behavior of C_l confirms that flow is full developed after ~ 5800 time steps.





Simulation of one period

2 Initializing the Simulation

For a clean simulation, we perform an additional 161 steps in order to finish with $C_l \simeq 0$. Then we proceed with 363 steps, which cover a complete period.

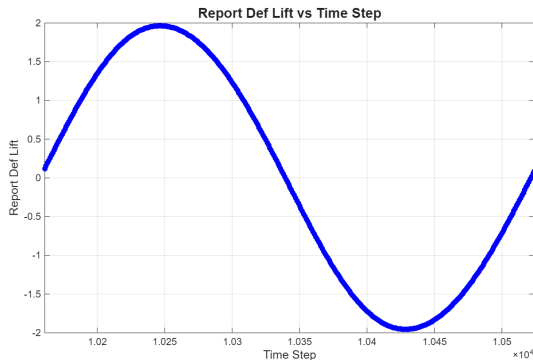


Figure: C_l values throughout one period.



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Retrieved Data

3 Results

After the simulation, we get the following data:

Reports
C_l
C_d
C_m
u on $y = 0$
u on $x = 0.5$

Statistics
$\overline{u_x}$ on $y=0$
$\overline{C_P}, \sigma(C_p)$
u, v statistics on $x = 0.5$



Contour plots

3 Results

We can take a look at the contour plot for the vorticity and k .

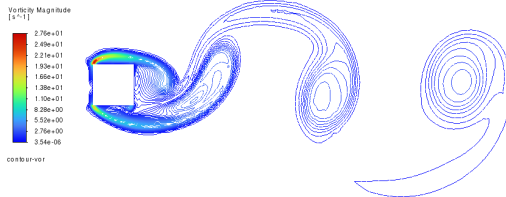


Figure: Vorticity contour plot for $k - \omega$ model at phase 1.

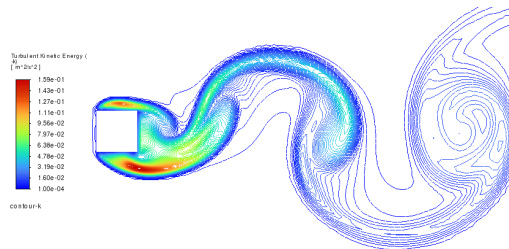


Figure: k contour plot for $k - \omega$ model at phase 1.



Integral Parameters

3 Results

Model/Integral Parameters	S_t	$\bar{C}_X$	$\tilde{C}_X$	$\tilde{C}_Y$
Lyn	0.135	2.05 - 2.23	n.a.	n.a.
Sakamoto	0.134	2.22	0.132	1.45
Durao	0.139	n.a.	n.a.	n.a.
Murakami	0.132	2.09	0.13	1.60
DIMAT	0.138	1.90	0.039	0.99
$k - \omega$	0.1374	2.02	0.141	1.393

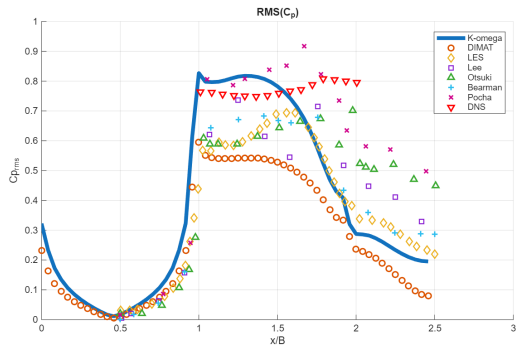
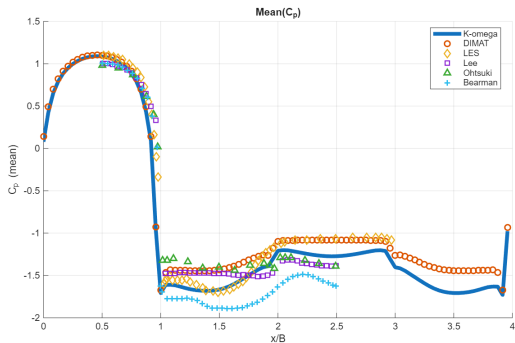
Table: Comparison of integral parameters of different turbulence models.



C_p statistics

3 Results

We can analyze and compare the data for $\overline{C_p}$ and $\text{RMS}(C_p)$.

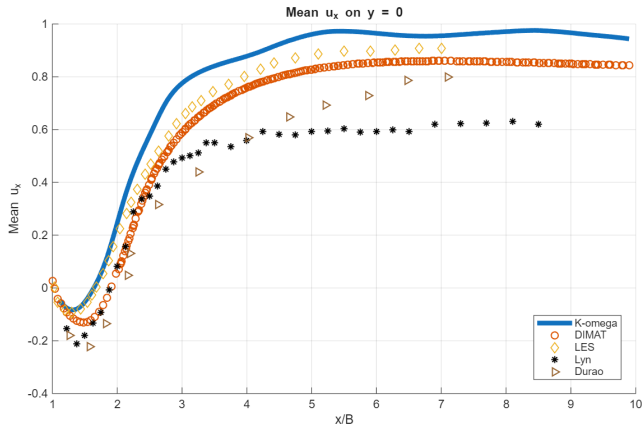




Horizontal velocity

3 Results

We can take a look at the mean horizontal velocity along $y = 0$.





Conclusions

3 Results

Comparison of $k - \omega$ model performance with respect to DIMAT:

	Mean C_p	RMS(C_p)	Mean u_x
$k - \omega$	Underpredicted	Overpredicted	Overpredicted

Overall, all values of our model are within the range of the other models.